

Mathematical Operation and Arithmetical Reasoning

Mathematical operation deals with a simple operators of math (Addition, subtraction, multiplication, division). These are the basic operation and also having less than, greater than, equal sign operators also.

The sign and symbols are given in the equation and we have to find out, the correct answer. There are two types of equations which are

- (i) Symbol based equations
- (ii) Equation checking

(i) Symbol based equations are having different types of letters, symbols which will represent any type of mathematical operations. We have to solve them in logical way and generate the answer.

Example : 100 M 25 G 2 K 12

M is shown for division G is shown for multiplication, K is shown for addition.

(ii) In equation checking, we have four equations and we have to check the correct equation on the basis of mathematical logic.

While solving, must remember the following rules :-

BODMAS

B → Bracket

O → Of

D → Division

M → Multiplication

A → Addition

S → Subtraction

Arithmetical Reasoning

In this arithmetical reasoning, there are given some problems based on ages, data based questions and other relative rules of the mathematics.

Solved Examples

1. If '+' means '-', '-' means 'x', 'x' means '÷' and '÷' means '+',

then,

$$25 \times 5 - 15 \div 3 + 4 = ?$$

Sol. $25 \div 5 \times 15 + 3 - 4$

$$= 5 \times 15 + 3 - 4 = 75 + 3 - 4 = 78 - 4 = 74$$

2. $11 * 7 * 9 * 2$ which option is correct ?

(a) $- \times \div$ (b) $+ = \times$ (c) $\times \div =$ (d) $+ = -$

Sol. Putting signs from option (2)

$$11 + 7 = 9 \times 2, 18 = 18$$

3. If '©' means 'subtraction', 'Δ' means 'addition', then find the value of $94 \Delta 27 \odot 44 \odot 56 \Delta 20$

(a) 41 (b) 45 (c) 47 (d) 48

Sol. Using proper symbols, we get the new sequence as

$$94 + 27 - 44 - 56 + 20$$

$$= 121 + 20 - 44 - 56 = 141 - 44 - 56$$

$$= 141 - 100 = 41$$

4. A student is typing on the keyboard of computer from 1 to 500 digits and numbers. Then how many times he will press the buttons of key-board ?

(a) 500 (b) 600 (c) 1000 (d) 1392

Sol. Digit, Number

$$1 - 9$$

$$10 - 99 \text{ (90 number)}$$

$$100 - 500 \text{ (401 number)}$$

∴ Total = $1203 + 180 + 9 = 1392$ time, he will press the buttons of key board.

5. There are some animals and labourers on a farm.

Total number of head are 120 and total number of feet are 340. Find the number of animals.

Sol. Let the no. of animals = x

$$\therefore \text{No. of labourers} = (120 - x)$$

$$\therefore 4x + 2(120 - x) = 340$$

$$4x + 240 - 2x = 340$$

$$2x = 100$$

$$x = 50$$

$$\therefore \text{No of animals} = 50$$

6. Your mother is 4 Years younger than your father. Your father is 6 times of your age. If your age is 6 years. Find the age of your mother

(a) 36 (b) 34 (c) 28 (d) 32

Sol. Father's age = $6 \times 6 = 36$ yr.

$$\therefore \text{Mother's age} = 36 - 4 = 32 \text{ yr.}$$

Practice Set

1. If ' $\times$ ' means ' $-$ ', ' $-$ ' means ' $\div$ ', ' $+$ ' means ' $\times$ ' and ' $\div$ ' means ' $+$ ', then what will be the value of the following expression?
 $16 - 8 + 4 \div 3 \times 9 = ?$
 (a) 10 (b) 19
 (c) 2 (d) 9
2. Some equations are solved on the basis of a certain system. On this basis, find out the correct answer from amongst the four alternatives for the unsolved equation:
 $A = 3(21) 247, B = 5(?) 407, C = 7(49) 567$
 (a) 28 (b) 25
 (c) 35 (d) 38
3. $7 \times 5 \times 4 = 57354, 8 \times 7 \times 3 = 78563$, then $6 \times 8 \times 5 = ?$
 (a) 86585 (b) 86855
 (c) 68485 (d) 86485
4. $a = 4(369) 9, b = 6(246) 4, c = 7(?) 3$
 (a) 303 (b) 213
 (c) 413 (d) 503
5. Which one of the following is incorrect?
 $6 * 3 * 4 * 5$
 (a) $\div, +, >$ (b) $\div, >, +$
 (c) $>, \div, +$ (d) $+, >, \div$
6. Some equations are solved on the basis of certain systems. On the same basis find out the correct answer for the unsolved equation.
 If $13 \times 12 = 651$ and $41 \times 23 = 448$, then $24 \times 22 = ?$
 (a) 504 (b) 46
 (c) 528 (d) 924
7. If $> = \div, \vee = \times, < = +, \wedge = -, + = =, \times = <, - = >$ then which of the following is correct.
 (a) $6 > 2 > 3 \wedge 8 \vee 4 + 13$
 (b) $6 \wedge 2 < 3 > 8 < 4 - 13$
 (c) $6 \vee 2 < 3 \wedge 8 > 4 \times 13$
 (d) $6 > 2 \vee 3 < 8 \wedge 4 + 13$
8. Which one of the following is correct?
 $96 * 6 * 8 * 2$
 (a) $\div, =, \times$ (b) $\times, =, \div$
 (c) $=, \div, \times$ (d) $=, \times, \div$
9. Some equations are solved on the basis of certain systems. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation in the question.
 $a = 12(390) 8, b = 7(134) 5, c = 5(?) 12$
 (a) 299 (b) 289
 (c) 279 (d) 280
10. Some equations are solved on the basis of a certain system. On the same basis, find the correct answer of the unsolved equation. If $837 = 452$ and $106 = 769$, then $708 \div 77 = ?$
 (a) 9 (b) 10
 (c) 11 (d) 12
11. Select correct combination (sequence) of mathematical signs to replace * signs to balance the equation:
 $9 * 4 * 22 * 14$
 (a) $\times = -$ (b) $\times - =$
 (c) $= - \times$ (d) $- \times =$
12. Which one of the following responses is correct?
 $8 * 5 * 27 * 3 * 16$
 (a) $\times, =, -, +$ (b) $-, =, \times, +$
 (c) $\times, =, +, -$ (d) $+, -, =, \times$
13. If X stands for addition, V stands for subtraction, U stands for equal to, Δ stands for division, Σ stands for multiplication, and σ stands for greater than and θ stands for less than. State which expression is true.
 (a) $3 X 8 V 2 U 12 \Delta 3$
 (b) $13 V 12 X 9 V 2 \sigma 5 \Sigma 1$
 (c) $2 \Sigma 3 \Sigma 4 \theta 51 \Delta 3$
 (d) $3 \Sigma 2 \Sigma 4 U 2 X 7 V 3$
14. Some equations are based on the basis of a certain system. Using the same solve the unsolved equation.
 If $10 - 3 = 12, 12 - 4 = 13, 14 - 5 = 14$, what is $16 - 6$?
 (a) 10 (b) 15
 (c) 16 (d) 18
15. After interchanging $\div$ and $\times$, 12 and 18, which one of the following equation becomes correct?
 (a) $(90 \times 18) \div 12 = 60$
 (b) $(18 \times 6) \div 12 = 2$
 (c) $(72 \div 18) \times 12 = 72$
 (d) $(12 \times 6) \div 18 = 36$
16. In the following question, Δ stands for any of the mathematical signs at different places, which are given as choices under each question. Select the choice with the correct sequence of signs which when substituted makes the question as a correct equation.
 $24 \Delta 4 \Delta 5 \Delta 4$
 (a) $\times + =$ (b) $= \times +$
 (c) $+ \times =$ (d) $= + \times$

17. The following equations follow a common property. Find out the correct value to complete D:
 $A = 51 (714) 14;$ $B = 61 (915) 15;$
 $C = 71 (1136) 16;$ $D = 81 (?) 17$
 (a) (1377) (b) (1378)
 (c) (1356) (d) (1346)
18. After interchanging $\div$ and $=$, 2 and 3 which one of the following statement becomes correct?
 (a) $15 = 2 \div 3$ (b) $5 \div 15 = 3$
 (b) $2 = 15 \div 3$ (d) $3 = 2 \div 15$
19. $25 * 2 * 6 = 4 * 11 * 0$
 Which set of symbols can replace $*$?
 (a) $\times, -, \times, +$ (b) $+, -, \times, +$
 (c) $\times, +, \times, -$ (d) $\times, +, +, \times$
20. If ' $\div$ ' stands for division ' $+$ ' stands for subtraction, ' $\times$ ' stands for multiplication, ' $\times$ ' stands for addition, then which one of the following equations is correct?
 (a) $70 - 2 + 4 \div 5 \times 6 = 44$
 (b) $70 - 2 + 4 \div 5 \times 6 = 21$
 (c) $70 - 2 + 4 \div 5 \times 6 = 341$
 (d) $70 - 2 + 4 \div 5 \times 6 = 36$
21. If $-$ stands for division, $+$ for multiplication, $\div$ for subtraction and $\times$ for addition, then which one of the following equation is correct?
 (a) $19 + 5 - 4 \times 2 \div 4 = 11$
 (b) $19 \times 5 - 4 \div 2 + 4 = 16$
 (c) $19 \div 5 + 4 - 2 \times 4 = 13$
 (d) $19 \div 5 + 4 + 2 \div 4 = 20$
22. If ' $\div$ ' stands for ' $\times$ ', ' $+$ ' stands for ' $\times$ ', ' $\times$ ' stands for ' $+$ ', and ' $\div$ ' stands for ' $\div$ ' which one of the following equation is correct?
 (a) $30 - 6 + 5 \times 4 \div 2 = 27$
 (b) $30 + 6 - 5 \div 4 \times 2 = 30$
 (c) $30 \times 6 \div 5 - 4 + 2 = 32$
 (d) $30 \div 6 \times 5 + 4 - 2 = 40$
23. If ' $\times$ ' means 'addition' ' $-$ ' means 'division', ' $\div$ ' means 'subtraction' and ' $+$ ' means 'multiplication', then which of the following equation is correct?
 (a) $16 + 5 - 10 \times 4 \div 3 = 9$
 (b) $16 - 5 \times 10 \div 4 + 3 = 12$
 (c) $16 + 5 \div 10 \times 4 - 3 = 9$
 (d) $16 \times 5 \div 10 \div 4 - 3 = 19$
24. If $+$, $-$, $\times$, $\div$, $=$, $>$ and $<$ are represented as δ , $\bullet$, γ , η , ω , β and α respectively, then which of the following is correct?
 (a) $3 \gamma 6 \eta 2 \delta 8 \bullet 4 \omega 5$
 (b) $3 \eta 6 \gamma 2 \delta 8 \bullet 4 \beta 5$
 (c) $3 \gamma 6 \bullet 2 \delta 8 \eta 4 \alpha 5$
 (d) $3 \delta 6 \bullet 2 \gamma 8 \eta 4 \omega 5$
25. If $\square = 6$, $\Delta = 3$, $\diamond = 5$, $\diamondsuit = 4$, $\square = 8$, $\square = 10$, then $(\square \times \Delta) \div \diamondsuit = ?$
 (a) $\square$ (b) $\diamond$
 (c) $\square$ (d) Δ
26. Among five friends A is shorter than B but taller than E, C is slightly taller than B but D is slightly shorter than B and slightly taller A. Who is the shortest?
 (a) A (b) E
 (c) C (d) D
27. Who is the shortest if
 (a) Sunitha is taller than Anitha.
 (b) Reena is taller than Chitra but shorter than Banu.
 (c) Anitha is shorter than Chitra.
 (d) Chitra is taller than Sunita
 (a) Chitra (b) Anitha
 (c) Reena (d) Banu
28. A man was 32 years of age when he had his first son. His wife was 35 years of age when his son attained the age of 7 years. The difference in age between the man and his wife is
 (1) 7 years (2) 3 years
 (3) 5 years (4) 4 years
29. A father's age is one more than 5 times of his son's age. After 3 years, the father's age would be 2 less than four times the son's age. Find the present age of the father.
 (a) 30 years (b) 40 years
 (c) 31 years (d) 29 years
30. My age is two years less than twice that of my brother. If I am sixteen years old, how old is my brother?
 (a) 7 years (b) 9 years
 (c) 10 years (d) 14 years
31. Hari is twice as old as Johnny, who is three years older than Rahul. If Hari's age is five times Rahul's age, how old is Johnny?
 (a) 2 years (b) 4 years
 (c) 5 years (d) 8 years
32. Hema was twice as old as Geetha 10 years ago. How old is Geetha today if Hema will be 40 years old 10 years hence?
 (a) 25 years (b) 20 years
 (c) 15 years (d) 35 years

33. Ashok's mother was 3 times as old as Ashok 5 years ago. After 5 years she will be twice as old as Ashok. How old is Ashok today ?
 (a) 10 years (b) 15 years
 (c) 20 years (d) 25 years
34. A man is 3 years older than his wife and four times as old as his son. If the son becomes 15 years old after 3 years, what is the present age of the wife ?
 (a) 48 (b) 52
 (c) 46 (d) 45
35. Naresh's age is 4 years less than twice the age of his brother. Which of the following represents the equation to find his age ?
 (a) $2x + 4$ (b) $4x + 2$
 (c) $x - 4$ (d) $2x - 4$
36. Shan is 55 years old, Sathian is 5 years junior to Shan and 6 years senior to Balan. The youngest brother of Balan is Devan and he is 7 years junior to him. So what is the age difference between Devan and Shan ?
 (a) 18 years (b) 15 years
 (c) 13 years (d) 7 years
37. Ravi has spent a quarter $\left(\frac{1}{4}\right)$ of his life as a boy, one-fifth $\left(\frac{1}{5}\right)$ as a youth, on-third $\left(\frac{1}{3}\right)$ as man and thirteen (13) years in old age. What is his present age ?
 (a) 70 years (b) 80 years
 (c) 60 years (d) 65 years
38. In a family, mother's age is twice that of daughter's age. Father is 10 years older than mother. Brother is 20 years younger than his mother and 5 years older than his sister. What is the age of the father ?
 (a) 62 years (b) 60 years
 (c) 58 years (d) 55 years
39. In a zoo, there are rabbits and pigeons. Their heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there?
 (a) 90 (b) 100
 (c) 110 (d) 120
40. A family, consist of a man, his wife, his three sons, their wives and three children in each son's family. How many members are there in the family.
 (a) 12 (b) 13
 (c) 15 (d) 17
41. There are 12 dozens of apples in a basket. Two dozens are added later. Ten apples got spoiled and are removed. The remaining are transferred equally into two baskets. How many are there in each.
 (a) 168 (b) 158
 (c) 79 (d) 89
42. In a family there are father, mother, four married sons & two unmarried daughters. Three sons have 2 daughters each and one has a son. How many female members are there in the family?
 (a) 13 (b) 8
 (c) 11 (d) 12
43. Exceeding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?
 (a) 9 (b) 10
 (c) 12 (d) 20
44. One third of Ramesh's marks in arithmetic is equal to half his marks in English. If he gets 150 marks in the two subjects together, how many marks has he got in English.
 (a) 60 (b) 120
 (c) 30 (d) 50
45. 14 notebook of a class were corrected with ink pen while 22 notebooks were corrected with colour pencil. If 4 notebooks were corrected with both, what is the strength of class?
 (a) 30 (b) 32
 (c) 28 (d) 25
46. In a class, there are 80 students who study both computer science and electronics. While 100 students study computer science, 120 study electronics. How many of them study computer science only?
 (a) 100 (b) 40
 (c) 180 (d) 20
47. Two bus tickets from city A to B and three tickets from city A to C cost Rs. 77 but three tickets from city A to B and two tickets from city A to C cost Rs. 73. What are the fares for cities B and C from A?
 (a) Rs. 4, 23 (b) Rs. 13, 17
 (c) Rs. 15, 14 (d) Rs. 17, 13
48. A bus leaves delhi with half the number of women as men. At meerut, 10 men get down and five women get in. Now there are equal number of men and women. How many passengers boarded the bus initially at delhi.

- (a) 36 (b) 45
(c) 15 (d) 30
49. A is 3 years older to B and 3 years younger to C while B and D are twins, How many years older is C to D?
(a) 2 (b) 3
(c) 6 (d) 12
50. A waiter's salary consists of his salary and tips. During one week his tips were $\frac{5}{4}$ of his salary. What fraction of this income came from tips.
(a) $\frac{4}{9}$ (b) $\frac{5}{4}$
(c) $\frac{5}{8}$ (d) $\frac{5}{9}$
- (a) 32 rolls (b) 54 rolls
(c) 108 rolls (d) 120 rolls
56. Some questions are solved on the basis of a certain system. Identify that operation and find out the correct answer from the four responses given.
A = 14 (290) 15, B = 16 (330) 17
C = 18 (?) 19
(a) 300 (b) 270
(c) 170 (d) 370
57. Find out the correct answer for the unsolved equation on the basis of the given equations.
If $6 * 5 = 91$
 $8 * 7 = 169$
 $10 * 7 = 211$
then $11 * 10 = ?$
(a) 331 (b) 993
(c) 678 (d) 845

Distinct Questions

51. A farmer built a fence around his square plot. He used 27 fence poles on each side of the square. How many poles did he need altogether?
(a) 100 (b) 104
(c) 108 (d) 105
52. Jack and Tom were travelling from town X to town Y which was 210 km apart. Jack set off 1 hour 15 minutes later than Tom but arrived 15 minutes earlier. If the average speed of Tom was 42 km/h. Find the average speed of Jack.
(a) 50 km/h (b) 60 km/h
(c) 65 km/h (d) 70 km/h
53. A car travels 20 miles in the same time as another car, travelling 20 MPH faster, covers 30 miles, How long does the journey take?
(a) 31 min (b) 29 min
(c) 30 min (d) 28 min
54. A college organised a annual function where $\frac{1}{5}$ of the girls and $\frac{1}{13}$ of the boys participated in the same. What are total number of students took part in the function.
(a) 15 (b) 45
(c) Date inadequate (d) 40
55. A tailor had a number of shirt pieces to cut from a roll of fabric. He cut each roll of equal length into 10 pieces. He cut at the rate of 45 cuts a minute. How many rolls would be cut in 24 minutes?
(a) 14400 (b) 15600
(c) 23040 (d) 17400
58. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.
If $324 \times 289 = 35$, $441 \times 484 = 43$,
 $625 \times 400 = 45$, find the value of 256×729 .
(a) 33 (b) 35
(c) 43 (d) 34
59. Some equations have been solved on the basis of certain system. Find the correct answer for the unsolved equation on that basis.
If $94 + 16 = 42$, $89 + 23 = 78$,
Then $63 + 45 = ?$
(a) 18 (b) 28
(c) 38 (d) 48
60. Which option is correct, according to some rule ?

2	3
4	5

(a) 14400 (b) 15600
(c) 23040 (d) 17400

Previous year Questions (Memory Based)

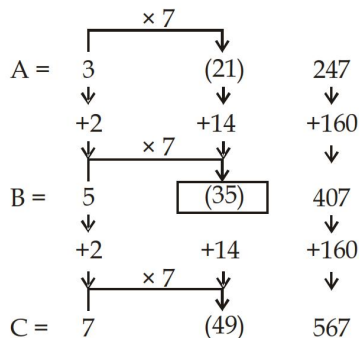
1. If '+' means '×', '-' means '÷', '×' means '-' and '÷' means '+' then what will be the value of $16 \div 64 - 8 \times 4 + 2$?
 (a) 12 (b) 24
 (c) 16 (d) 18
2. If $+=\times$, $=\div$, $\times=+$, $\div=-$, then which is the correct equation out of the following.
 (a) $18 + 6 - 4 \times 2 \div 3 = 26$ (b) $18 \div 6 + 4 - 2 \div 3 = 22$
 (c) $18 \times 6 - 4 + 7 \times 8 = 47$ (d) $18 - 6 \times 7 + 8 = 63$
3. If + stands for division, × stand for addition, - stands for multiplication, ÷ stands for subtraction, which of the following is correct?
 (a) $46 \times 6 \div 4 - 5 + 3 = 74$
 (b) $46 - 6 + 4 \times 5 \div 3 = 71$
 (c) $46 \div 6 \times 4 - 5 + 3 = 75.5$
 (d) $46 \times 6 - 4 + 5 \div 3 = 70.1$
4. Some equations have been solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.
 If $9 * 7 = 32$, $13 * 7 = 120$, $17 * 9 = 208$, then $19 * 11 = ?$
 (a) 150 (b) 180
 (b) 210 (d) 240
5. If '+' means '-', '-' means '×', '×' mean '÷' and '÷' means '+' which combination will give you the value of 0 in $48 * 4 * 12 * 20 * 4$?
 (a) $+-\div\times$ (b) $\div+-\times$
 (c) $-\div\times+$ (d) $\times\div++$
6. After interchanging + and -, 8 and 7, which one of the following become correct
 (a) $6 + 8 \times 2 - 7 = 0$ (b) $8 \times 2 + 7 - 6 = 9$
 (c) $7 \times 8 + 6 - 9 = 25$ (d) $8 - 7 + 3 \times 5 = 35$
7. John's age is 42 years and Kelvin's age is 26 years. How many years ago was Kelvin's age half of John's age?
 (a) 4 years (b) 10 years
 (c) 8 years (d) 6 years
8. Select the correct combination of mathematical signs to replace * signs and to balance the following equation.
 $(\sqrt{121} * 9) * 5 * 1 * 4$
 (a) $+, \div, \times =$ (b) $=, +, \times, \div$
 (c) $-, \times, + =$ (d) $-, +, \times =$
9. If '-' stands for 'division', '+' stands for 'multiplication', ÷ stands for 'subtraction', '×' stands for 'addition', then which one of the following equations is correct?
 (a) $52 \div 4 + 5 \times 8 - 2 = 36$ (b) $36 - 12 \times 6 \div 3 + 4 = 60$
 (c) $43 \times 7 \div 5 + 4 - 8 = 25$ (d) $36 \times 4 - 12 + 5 \div 3 = 420$
10. Which of the following interchange of signs would make the equation correct?
 $49 - 7 + 7 = 14$
 (a) - and + (b) - and +
 (c) ÷ and + (d) + and -
11. The ratio of the present ages of Sunita and Vinita is 4 : 5, six years hence, the ratio of their ages will be 14 : 17. What will be the ratio of their ages 12 years hence.
 (a) 13 : 15 (b) 16 : 19
 (c) 17 : 19 (d) 15 : 19
12. Select the correct combination of mathematical signs to replace * signs and to balance the following equation:
 $(8 * 5 * 6) * 10 * 24$
 (a) $+-=\times$ (b) $\times-\div=$
 (c) $-\div+=$ (d) $\times\times\div=$
13. If L denotes ×, M denotes ÷, P denotes + and Q denotes -, the find the value of
 $16 P 24 M 8 Q 6 M 2 L 3 = ?$
 (a) 6 (b) 8
 (c) 10 (d) 12
14. X stands for +, Z stands for +, Y stands for -, and P stands for ×, then what is the value of $10 P 2 X 5 Y 5$?
 (a) 10 (b) 15
 (c) 20 (d) 25
15. Select the correct combination of mathematical signs to replace * signs and to balance the given equation:
 $28 * 4 * 9 * 16$
 (a) $\div + =$ (b) $+ \div =$
 (c) $- \times +$ (d) $- = \times$
16. If × stands for +, ÷ stands for -, - stands for ÷, + stands for ×, which one of the following is correct?
 (a) $11 - 6 \div 4 + 2 = 64$
 (b) $10 - 5 \times 6 \div 2 = 54$
 (c) $15 + 3 \times 4 \div 3 = 12$
 (d) $12 + 4 \div 280 - 7 = 8$
17. If L denotes ×, M denotes ÷, P denotes +, Q denotes -
 Then $16 P 24 M 8 Q 6 M 2 L 3 = ?$
 (a) 10 (b) 9
 (c) 12 (d) 11
18. Select the correct set which will fit in the given equation
 $8 * 8 * 1 * 11 * 11$
 (a) $+=\div-$ (b) $\times+=\div$
 (c) $\div\times+=$ (d) $-+=\div$
19. $7 * 5 * 5 * 4 * 10$
 (a) $+\div=-$ (b) $\times-=\times$
 (c) $\times+=\times$ (d) $+ \times \div =$

20. If 'P' means '+', 'Q' means '×', 'R' means '÷' and 'S' means '-' then-
 $44 Q 9 R 12 S 6 Q 4 P 16 = ?$
 (a) 25 (b) 112
 (c) 36 (d) 124
21. There are some ducks and goats in a ground. Total no of heads and feet are 77 and 224 respectively. How many ducks are there in the ground?
 (a) 42 (b) 30
 (c) 32 (d) 47
22. If all the numbers from 501 to 700 are written, what is the total number of times does the digit 6 appear
 (a) 138 (b) 139
 (c) 140 (d) 141
23. A book has 300 pages and each page has 20 lines of 10 – 10 words. Find the no. of words in the book.
 (a) 6000 (b) 60,000
 (c) 66000 (d) 600000
24. Some oranges were distributed among 40 children equally. If 20 more children were there, each children would have been get 5 less oranges. How many oranges are there at the beginning?
 (a) 200 (b) 300
 (c) 400 (d) 600
25. A shepherd had 17 sheep. All but nine died. How many was he left with?
 (a) 0 (b) 8
 (c) 9 (d) 17
26. In a exam, a student gets 4 marks for each correct answer, but loses one for a wrong answer, There were 60 questions and got 130 marks. Find the correct number of answer.
 (a) 42 (b) 35
 (c) 38 (d) 40
27. In each 1 km distance a bike is standing. Then how many bikes are there in 10 km distance?
 (a) 10 (b) 9
 (c) 11 (d) 12
28. The number of boys in a class is three times the number of girls, which one of the following numbers cannot represent the total number of children in the class?
 (a) 48 (b) 44
 (c) 42 (d) 40
29. what is the sum of numbers from 1 to 5000?
 (a) 1250500 (b) 12505200
 (c) 1205500 (d) 12502500
30. How many words with or without meaning can be made using the letters of word 'INDIA'?
 (a) 40 (b) 60
 (c) 80 (d) 45

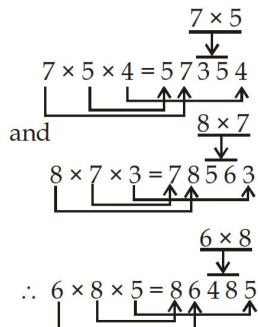
Practice Set Solutions

1. (c); $16 \div 8 \times 4 + 3 - 9 \Rightarrow 8 + 3 - 9 = 2$

2. (c);



3. (d);



4. (b); $4 \times 9 = 36$; $3 + 6 = 9 \Rightarrow 369$

$6 \times 4 = 24$; $2 + 4 = 6 \Rightarrow 246$

$7 \times 3 = 21$; $1 + 2 = 3 \Rightarrow 213$

5. (b); $6 \div 3 > 4 + 5$

$\rightarrow 2 > 9$ in correct

6. (d); $13 \times 12 \Rightarrow 31 \times 21 = 651$

$41 \times 23 \Rightarrow 14 \times 32 = 448$

Similarly,

$24 \times 22 \Rightarrow 42 \times 22 = 924$

7. (d); $6 > 2 > 3 \wedge 8 \vee 4 + 13$

$\Rightarrow 6 \div 2 \div 3 - 8 \times 4 = 13$

$\Rightarrow 1 - 32 \neq 13$

$6 \wedge 2 < 3 > 8 < 4 - 13$

$\Rightarrow 6 - 2 + \frac{3}{8} + 4 > 13$

$\Rightarrow \frac{48 - 16 + 3 + 32}{8} > 13$

$\Rightarrow \frac{67}{8} > 13$

$6 \vee 2 < 3 \wedge 8 > 4 \times 13$

$$\Rightarrow 6 \times 2 + 3 - 8 \div 4 < 13$$

$$\Rightarrow 12 + 3 - 2 < 13$$

$$\Rightarrow 13 < 13$$

$$6 > 2 \vee 3 < 8 \wedge 4 + 13$$

$$\Rightarrow 6 \div 2 \times 3 + 8 - 4 = 13$$

$$\Rightarrow 3 \times 3 + 8 - 4 = 13$$

$$\Rightarrow 9 + 8 - 4 = 13$$

8. (a); $96 \div 6 = 8 \times 2$

$$\Rightarrow 16 = 16$$

9. (c); $a = 12$ (390) $8 \Rightarrow 12 + 8 = 20$;

$$20 \times 20 - 10 = 390$$

$$b = 7$$
 (134) $5 \Rightarrow 7 + 5 = 12$;

$$12 \times 12 - 10 = 134$$

$$c \Rightarrow 5 + 12 = 17; 17 \times 17 - 10$$

$$= 289 - 10 = 279$$

10. (d); $\begin{array}{ccc} 8 & 3 & 7 \\ \downarrow & \downarrow & \downarrow \end{array}$

$$\begin{array}{ccc} 4 & 5 & 2 \\ \downarrow & \downarrow & \downarrow \end{array}$$

$$\begin{array}{ccc} 7 & 0 & 8 \\ \downarrow & \downarrow & \downarrow \end{array}$$

$$\begin{array}{ccc} 2 & 6 & 4 \end{array}$$
 and $\begin{array}{cc} 2 & 2 \\ \downarrow & \downarrow \end{array}$

$$2 \ 6 \ 4 \text{ and } 2 \ 2$$

$$\text{Now, } 264 \div 22 = 12$$

11. (b); $9 \times 4 - 22$

$$36 - 22 = 14$$

12. (a); $8 \times 5 = 27 - 3 + 16$

$$\Rightarrow 40 = 43 - 3$$

$$\Rightarrow 40 = 40$$

13. (b);

$X \Rightarrow +$	$V \Rightarrow -$	$U \Rightarrow =$	$\theta \Rightarrow <$
$\wedge \Rightarrow \div$	$\Sigma \Rightarrow \times$	$\sigma \Rightarrow >$	

$$3 \times 8 \vee 2 \cup 12 \wedge 3$$

$$\Rightarrow 3 + 8 - 2 = 12 \div 3$$

$$\Rightarrow 9 \neq 4$$

$$13 \vee 12 \times 9 \vee 2 \sigma 5 \Sigma 1$$

$$\Rightarrow 13 - 12 + 9 - 2 > 5 \times 1$$

$$\Rightarrow 8 > 5$$

$$2 \Sigma 3 \Sigma 4 \sigma 5 1 \wedge 3$$

$$\Rightarrow 2 \times 3 \times 4 < 5 1 \div 3$$

$$\Rightarrow 24 \neq 17$$

$$3 \Sigma 2 \Sigma 4 \cup 2 \times 7 \vee 3$$

$$3 \times 2 \times 4 = 2 + 7 - 3$$

$$\Rightarrow 24 \neq 6$$

14. (b); $10 - 3 = 7; 7 + 5 = 12$

$$12 - 4 = 8; 8 + 5 = 13$$

$$14 - 5 = 9; 9 + 5 = 14$$

$$16 - 6 = 10; 10 + 5 = 15$$

15. (d); $(18 \div 6) \times 12 = 36$

16. (b,d); Option (b)

$$24 = 4 \times 5 + 4$$

$$\Rightarrow 24 = 20 + 4$$

Option (d)

$$24 = 4 + 5 \times 4$$

$$\Rightarrow 24 = 4 + 20$$

Both options (b) and (d) are correct.

17. (a); $A = 51 \times 14 = 714$

$$B = 61 \times 15 = 915$$

$$C = 71 \times 16 = 1136$$

$$D = 81 \times 17 = 1377$$

18. (b); $5 = 15 \div 3$

19. (a); $25 \times 2 - 6 = 4 \times 11 + 0$

$$\Rightarrow 50 - 6 = 44 + 0 \Rightarrow 44 = 44$$

20. (b); $70 \div 2 - 4 \times 5 + 6 = 44$

$$\Rightarrow 35 - 20 + 6 = 44$$

$$\Rightarrow 15 + 6 \neq 44$$

$$70 \div 2 - 4 \times 5 + 6 = 21$$

$$\Rightarrow 35 - 20 + 6 = 21$$

$$\Rightarrow 41 - 20 = 21$$

21. (c);

$- \Rightarrow \div$	$+ \Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow +$

$$19 \times 5 \div 4 + 2 - 4 = 11$$

$$\Rightarrow \frac{19 \times 5}{4} + 2 - 4 \neq 11$$

$$19 + 5 \div 4 - 2 \times 4 = 16$$

$$\Rightarrow 19 + \frac{5}{4} - 8 = 16$$

$$\Rightarrow 76 + 5 - 32 = 16 \times 4$$

$$\Rightarrow 81 - 32 \neq 64$$

$$\Rightarrow 19 - 5 \times 4 \div 2 + 4 = 13$$

$$\Rightarrow 19 - 5 \times 2 + 4 = 13$$

$$\Rightarrow 19 - 10 + 4 = 13$$

$$19 - 5 \times 4 \times 2 - 4 = 20$$

$$\Rightarrow 19 - 40 - 4 \neq 20$$

22. (a); $30 - 6 + 5 \times 4 \div 2 = 27$

$$\Rightarrow 30 \div 6 \times 5 + 4 - 2 = 27$$

$$\Rightarrow 25 + 4 - 2 = 27$$

$$30 + 6 - 5 \div 4 \times 2 = 30$$

$$\Rightarrow 30 \times 6 \div 5 - 4 + 2 = 30$$

$$\Rightarrow 36 - 4 + 2 \neq 30$$

$$30 \times 6 \div 5 - 4 + 2 = 32$$

$$\Rightarrow 30 + 6 - 5 \div 4 \times 2 \neq 32$$

$$30 \div 6 \times 5 + 4 - 2 = 40$$

$$30 - 6 + 5 \times 4 \div 2 = 40$$

$$\Rightarrow 30 - 6 + 10 \neq 40$$

23. (a); $16 + 5 - 10 \times 4 \div 3 = 9$
 $\Rightarrow 16 \times 5 \div 10 + 4 - 3 = 9$
 $\Rightarrow 8 + 4 - 3 = 9$

24. (d);

$+\Rightarrow\delta$	$-\Rightarrow\bullet$	$\times\Rightarrow\gamma$	$\div\Rightarrow\eta$
$=\Rightarrow w$	$>\Rightarrow\beta$	$<\Rightarrow\alpha$	

Option (1)

$3\gamma 6\eta 2\delta 8\bullet 4\omega 5$
 $\Rightarrow 3 \times 6 \div 2 + 8 - 4 = 5$
 $\Rightarrow 3 \times 3 + 8 - 4 = 5$
 $\Rightarrow 17 - 4 \neq 5$

Option (2)

$3\eta 6\gamma 2\delta 8\bullet 4\beta 5$
 $\Rightarrow 3 \div 6 \times 2 + 8 - 4 > 5$
 $\Rightarrow \frac{3}{6} \times 2 + 8 - 4 > 5$
 $\Rightarrow 5 \neq 5$

Option (3)

$3\gamma 6\bullet 2\delta 8\eta 4\alpha 5$
 $\Rightarrow 3 \times 6 - 2 + 8 \div 4 < 5$
 $\Rightarrow 3 \times 6 - 2 + 2 < 5$
 $\Rightarrow 18 - 2 + 2 \neq 5$

Option (4)

$3\delta 6\bullet 2\gamma 8\eta 4\omega 5$
 $\Rightarrow 3 + 6 - 2 \times 8 \div 4 = 5$
 $\Rightarrow 3 + 6 - 2 \times 2 = 5$
 $\Rightarrow 9 - 4 = 5$

25. (a); $(\square \times \Delta) \div \diamond = ?$
 $\Rightarrow ? = (8 \times 3) \div 4$
 $\Rightarrow ? = 24 \div 4 = 6 \Rightarrow \square$

26. (b); $B > A > E$
 $C > B, B > D > A$
 $\therefore C > B > D > A > E$

27. (b); Chitra > Sunitha > Anitha ... (i)
 Banu > Reena > Chitra ... (ii)
 From statements (i) and (ii)
 Anitha is the shortest.

28. (d); At the birth of son, the age of mother = $35 - 7 = 28$ years
 Difference in age
 $= 32 - 28 = 4$ years

29. (c); Suppose the age of son is x years
 Therefore, the age of father will be $5x + 1$
 Again, $4(x + 3) - 2 = 5x + 1 + 3$
 $\Rightarrow 4x + 12 - 2 = 5x + 4$

$\Rightarrow 10 - 4 = 5x - 4x$
 $\therefore x = 6$
 Age of father = $5x + 1$
 $= 5 \times 6 + 1 = 31$ years

30. (b); According to question
 $2x - 2 = 16 \Rightarrow 2x = 16 + 2$
 $\therefore x = 9$ years

31. (c); Suppose the age of Johnny is x years and that of Rahul is y years.
 According to question
 Age of Hari = $2x = 5y$
 or, $2x - 5y = 0$... (i)
 Again $x = 3 + y$
 or, $x - y = 3$... (ii)
 On solving equations (i) and (ii), we get
 $y = 2$

$\therefore$ Age of Johnny = $3 + y = 3 + 2 = 5$ years

32. (b); Suppose the present age of Geetha is x years
 Present age of Hema
 $= 40 - 10 = 30$ years
 According to question
 $2(x - 10) = 30 - 10$
 $\Rightarrow 2x - 20 = 20$
 $\Rightarrow 2x = 20 + 20 = 40$
 $\therefore x = \frac{40}{2} = 20$ years

33. (b); Suppose the present age of Ashok is x years and that of his mother is y years.
 5 years ago
 $3(x - 5) = (y - 5)$
 $\Rightarrow 3x - 15 = y - 5$
 $\Rightarrow 3x - y = 10$... (i)
 5 years hence.
 $2(x + 5) = (y + 5)$
 $\Rightarrow 2x + 10 = y + 5$
 $\Rightarrow 2x - y = -5$... (ii)
 From equations (i) and (ii)
 $x = 15$ years.

34. (d); Suppose the present age of son is x years.
 Therefore, present age of the father = $4x$ years.
 According to question
 $x + 3 = 15$
 $\therefore x = 15 - 3 = 12$
 The present age of Father = $4x = 4 \times 12 = 48$
 $\therefore$ The present age of man's wife
 $= 48 - 3 = 45$ years

35. (d); Suppose the age of Naresh's brother = x years
 $\therefore$ Age of Naresh = $2x - 4$
36. (a); Age of Shan = 55 years
 Age of Sathian = $55 - 5 = 50$ years
 Age of Balan = $50 - 6 = 44$ years
 Age of Devan = $44 - 7 = 37$ years
 Difference between the ages of Shan and Devan
 $= 55 - 37 = 18$ years
37. (c); Suppose his present age is x years
 According to question

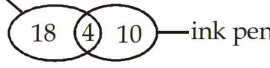
$$\frac{x}{4} + \frac{x}{5} + \frac{x}{3} = x - 13$$

$$\Rightarrow \frac{15x + 12x + 20x}{60} = x - 13$$

$$\Rightarrow 47x = 60x - 780 \Rightarrow 60x - 47x = 780$$

$$\Rightarrow 13x = 780$$

$$\therefore x = \frac{780}{13} = 60 \text{ years}$$
38. (b); Suppose the age of daughter is x years.
 Age of brother = $x + 5$
 Age of mother = $2x$ years
 $\therefore 2x - 20 = x + 5$
 $\Rightarrow 2x - x = 5 + 20$
 $\Rightarrow x = 25$ years
 Age of mother = $2x = 2 \times 25 = 50$ years
 Age of father = $50 + 10 = 60$ years
39. (c); Suppose the number of rabbits = x
 Number of pigeons = y .
 According to question,
 $x + y = 200$... (i)
 and $4x + 2y = 580$... (ii)
 From equation (i) and (ii)
 $x = 90$
 $y = 200 - 90 = 110$
40. (d); The man and his wife = 2 members
 Three sons and their wives = 6 members
 Three children each of three sons = $3 \times 3 = 9$ members.
 Total number of candidates = $2 + 6 + 9 = 17$.
41. (c); Total no. of apples = $14 \times 12 = 168$
 Ten apples were removed, remaining apples
 $= 168 - 10 = 158$
 Now $\frac{158}{2} = 79$.

42. (a); Mother = 1, wives = 4, unmarried daughters = 2
 Daughters = $3 \times 2 = 6$.
 Total number of female members
 $1 + 4 + 2 + 6 = 13$
43. (b); Relative speed $(54 - 45) = 9$ kmph
 Stoppage per hour = $\frac{9}{54} \times 60 = 10$ minutes
44. (a); $\frac{\text{Arithmetic marks}}{3} = \frac{\text{English}}{2}$
 $\Rightarrow 2A - 3E = 0$... (i)
 $\Rightarrow A + E = 150$... (ii)
 From equation (i) and (ii) $E = 60$
45. (b); Colour Pencil

 Total Strength = $18 + 4 + 10 = 32$
46. (d); There are 100 students in computer science which include students of electronics too.
 Number of students opting for computer science only
 $= 100 - 80 = 20$.
47. (b); Fare of city B from city A = Rs. x
 Fare of city C from city A = Rs. y
 Then $2x + 3y = 77$... (i)
 $3x + 2y = 73$... (ii)
 By solving equation (i) and (ii) $y = 17$, then $x = 13$.
48. (b); Suppose the number of women boarded the bus at delhi is x . Therefore then number of men = $2x$.
 According to the question
 $2x - 10 = x + 5$
 $\Rightarrow x = 15$
 Total number of passengers boarded the bus initially = $3x = 3 \times 15 = 45$
49. (c); Since B and D are twins, so $B = D$
 Now, $A = B + 3$ and $A = C - 3$
 Thus $B + 3 = C - 3$
 $D + 3 = C - 3$
 $C - D = 6$
50. (d); Let salary = x Rs. Then tips $\frac{5}{4}x$
 Total income = Rs. $\left(x + \frac{5}{4}x\right) = \frac{9x}{4}$
 Required fraction = $\frac{5x}{4} \times \frac{4}{9x} = \frac{5}{9}$

Distinct Solutions

51. (b); Since each pole at the corner of the plot is common to its two sides, so we have total number of poles needed
 $= 27 \times 4 - 4 = 108 - 4 = 104$
52. (b); Tom covered distance in $\frac{210}{42} = 5$ hours.
 Time taken by jack to cover 210 km
 $= 5.00 - 1.30 = 3.30$ hours $= 3\frac{1}{2}$ hr.
 $\therefore$ Average speed of jack $= \frac{210}{\frac{7}{2}} = \frac{210 \times 2}{7} = 60$ km/h
53. (c); Difference in the distance covered by both the cars
 $= (30 - 20)$ miles $= 10$ miles.
 The faster car travelling 20 MPH faster, covers 10 miles more,. Therefore, the time of journey $= \frac{10}{20} = \frac{1}{2}$ hours $= 30$ min.
54. (c); These fraction of boys and girl are dependent on a digit which is not given in question, so given data is inadequate.
55. (d); Number of cuts made to cut a roll into 10 pieces = 9
 Therefore required number of rolls
 $= (45 \times 24)/9 = 120$

56. (d); $(14 + 15) \times 10$
 $= 29 \times 10 = 290$
 $(16 + 17) \times 10 = 33 \times 10 = 330$
 Therefore,
 $(18 + 19) \times 10 = 37 \times 10 = 370$
57. (a); $6 \times 5 = 30$
 $30 \times 3 = 91$
 $8 \times 7 = 56$
 $56 \times 3 + 1 = 169$
 $10 \times 7 = 70$
 $70 \times 3 + 1 = 211$
 Similarly,
 $11 \times 10 = 110$
 $110 \times 3 + 1 = 331$
58. (c); $\sqrt{324} = 18$; $\sqrt{289} = 17$
 $18 + 17 = 35$
 $\sqrt{441} = 21$; $\sqrt{484} = 22$
 $21 + 22 = 43$
 $\sqrt{625} = 25$; $\sqrt{400} = 20$
 $25 + 20 = 45$
 $\sqrt{256} = 16$; $\sqrt{729} = 27$
 $16 + 27 = 43$
59. (c); $9 \times 4 + 1 \times 6 = 36 + 6 = 42$
 $8 \times 9 + 2 \times 3 = 72 + 6 = 78$
 Similarly
 $6 \times 3 + 4 \times 5 = 18 + 20 = 38$
60. (a); $2^2 \times 3^2 \times 4^2 \times 5^2 = 14400$

Previous year Solutions (Memory Based)

1. (c); $16 \div 64 - 8 \times 4 + 2$
 $\Rightarrow 16 \div 64 \div 8 - 4 \times 2$
 $\Rightarrow 16 \div 8 - 8$
 $\Rightarrow 16$
2. (a); $18 + 6 - 4 \times 2 \div 3 = 26$
 $\Rightarrow 18 \div 6 \div 4 + 2 - 3$
 $\Rightarrow 27 + 2 - 3$
 $\Rightarrow 26$
3. (b); $46 - 6 + 4 \times 5 \div 3 = 71$
 $\Rightarrow 46 \div 6 \div 4 + 5 - 3$
 $\Rightarrow 69 + 5 - 3$
 $\Rightarrow 71$
4. (d); $9 + 7 = 16$; $9 - 7 = 2$
 $16 \times 2 = 32$
 $13 + 7 = 20$; $13 - 7 = 6$
 $20 \times 6 = 120$
5. (d); $48 \times 4 \div 12 + 20 + 4$
 $\Rightarrow 48 \div 4 + 12 - 20 - 4$
 $\Rightarrow 12 + 12 - 20 - 4$
 $\Rightarrow 24 - 24 = 0$
6. (a); $6 + 8 \times 2 - 7 = 0$
 $\Rightarrow 6 - 7 \times 2 + 8 = 0$
 $\Rightarrow 6 - 14 + 8 = 0$
7. (b); Suppose x years ago, kelvin's age was half of John's then
 $\frac{42-x}{2} = 26 - x$, $42 - x = 52 - 2x$
 $x = 10$ year

8. (a); $\sqrt{121} + 9 \div 5 \times 1 = 4$

$$(11 + 9) \div 5 = 4$$

$$20 \div 5 = 4 \Rightarrow 4 = 4$$

9. (a); $52 \div 4 + 5 \times 8 - 2 = 36$

$$52 - 4 \times 5 + 8 \div 2 = 36$$

$$52 - 20 + 4 = 36$$

$$56 - 20 = 36$$

$$36 = 36$$

10. (c); $49 - 7 + 7 = 14$

$$\Rightarrow 49 \div 7 + 7 = 14$$

$$\Rightarrow 7 + 7 = 14$$

11. (b); Present ages ratio is 4 : 5

$$\text{After 6 years} = \frac{4x+6}{5x+6} = \frac{14}{17}$$

$$17(4x+6) = 14(5x+6)$$

$$68x + 102 = 70x + 84$$

$$2x = 18, x = 9$$

So present age of Sunita and Vinita is 36 & 45 respectively

$$\text{After 12 years} = \frac{48}{57} = \frac{16}{19}$$

12. (d); $(8 \times 5 \times 6) \div 10 = 24$

$$240 \div 10 = 24, 24 = 24$$

13. (c); 16 P 24 M 8 Q 6 M 2 L 3

$$\Rightarrow 16 + 24 \div 8 - 6 \div 2 \times 3$$

$$\Rightarrow 16 + 3 - 3 \times 3$$

$$\Rightarrow 19 - 9$$

$$\Rightarrow 10$$

14. (c); 10 P 2 X 5 Y 5

$$\Rightarrow 10 \times 2 + 5 - 5$$

$$\Rightarrow 20 + 5 - 5$$

$$\Rightarrow 25 - 5$$

$$\Rightarrow 20$$

15. (a) $28 * 4 * 9 * 16$

$$\Rightarrow 28 \div 4 + 9 = 16$$

$$\Rightarrow 7 + 9 = 16$$

$$\Rightarrow 16 = 16$$

16. (d); $11 - 6 \div 4 + 2 = 64$

$$11 \div 6 - 4 \times 2 = 64$$

$$11 \div 6 - 8 \neq 64$$

$$1.83 - 8 \neq 64$$

$$12 + 4 \div 280 - 7 = 8$$

$$12 \times 4 - 280 \div 7 = 8$$

$$48 - 40 = 8, 8 = 8$$

17. (a); $16 + 24 \div 8 - 6 \div 2 \times 3$

$$\Rightarrow 16 + 3 - 3 \times 3$$

$$\Rightarrow 16 + 3 - 9 \Rightarrow 16 - 6 = 10$$

18. (d); $8 - 8 + 1 = 11 \div 11 = 1 = 1$

19. (c); $7 \times 5 + 5 = 4 \times 10$

$$35 + 5 = 40, 40 = 40$$

20. (a); $44 \times 9 \div 12 - 6 \times 4 + 16$

$$44 \times \frac{9}{12} - 6 \times 4 + 16$$

$$33 - 24 + 16 = 49 - 24 = 25$$

21. (a); Total number of head = 77

$$\text{Step I: } 77 \times 4 = 308$$

$$\text{Step II: } 308 - 224 = 84$$

$$\text{Step III: No. of ducks} = \frac{84}{2} = 42$$

$$\text{No of goats} = 77 - 42 = 35$$

22. (c); No. of 6 b/w 501 to 599 at tens place = 10

$$\text{No. of 6 b/w 501 to 599 at unit place} = 10$$

$$\text{No. of 6 b/w 600 to 699 at hundred place} = 100$$

$$\text{No. of 6 b/w 601 to 699 at tens place} = 10$$

$$\text{No. of 6 b/w 601 to 699 at unit place} = 10$$

$$\therefore \text{Total} = 10 + 10 + 100 + 10 + 10 = 140$$

23. (b); Total no. of words = $300 \times 10 \times 20 = 60,000$

24. (d); Let total no. of oranges = x

$$\text{and Number of oranges given to each children} = y$$

$$\therefore 40y = x \quad \dots (i)$$

$$\text{Again, } 60(y - 5) = x$$

$$60y - 300 = x \quad \dots (ii)$$

$$\text{from (i) and (ii)}$$

$$40y = 60y - 300$$

$$y = 15$$

$$\therefore \text{No. of oranges (x)} = 40 \times 15 = 600$$

25. (c); 'All but nine died' means 'All except nine died i.e. nine sheep remained alive.

26. (c); Let the no. of correct answer = x

$$\text{Then no. of wrong answer} = 60 - x$$

$$\text{A.T.Q.} \Rightarrow 4x - (60 - x) = 130$$

$$5x - 60 = 130$$

$$5x = 190, x = 38$$

27. (c); Total bike = $10 + 1 = 11$

28. (c); Boys: Girls = 3 : 1

$$42, \text{ is not possible as it is not divisible by } (3 + 1) = 4$$

29. (d); $\text{Sum} = \frac{n(n+1)}{2} = \frac{5000 \times (5000 + 1)}{2}$

$$= 2500 \times 5001 = 12502500$$

30. (b); No. of words = $\frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60$